

An Agent-Based Model for *Aedes aegypti* Population Dynamics with Spatial Heterogeneity

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Mosquitoes, particularly ***Aedes aegypti***, transmit diseases such as dengue, Zika, and chikungunya, making the estimation of mosquito population density important for vector control. This work extends a previous study [3] by incorporating spatial heterogeneity environments and stochastic interaction rules inspired by reaction–diffusion models [4]. Population dynamics are represented by interacting agents evolving on a discrete spatial lattice. A subset $\mathcal{S} = \{s_1, \dots, s_m\} \subset \mathcal{G}$ defines breeding sites where juveniles develop. Reflective boundaries keep agents inside the domain.

Mosquitoes are modeled as agents with five states: juveniles (J), adult males (M), unmated females (F_U), gravid females (F_G), and dead (D), inspired by [2]. Each agent at time t is defined as $a_i(t) = (s_i(t), x_i(t))$, where $s_i \in \{J, M, F_U, F_G, D\}$ and $x_i \in \mathcal{G}$. Local interactions within a finite radius determine density-dependent mortality, mating encounters, and oviposition events. Biological parameters include juvenile and adult mortality rates μ_J, μ_M, μ_F , maturation rate γ , larval competition α with carrying capacity K_c , mating rate β , and oviposition rate f . Spatial interactions occur within radius r_m, r_d , and r_o . Juveniles exist only at breeding sites. Let $J_k(t)$ denote the number of juveniles associated with breeding site s_k . Their evolution follows

$$J_k(t + \Delta t) = J_k(t) + X_{\text{birth},k} - X_{\text{maturation},k} - X_{\text{death},k}, \quad (1)$$

where the stochastic transitions follow binomial processes $X_{\text{birth},k} \sim \text{Bin}\left(F_G^{(k)}, 1 - e^{-f\Delta t}\right)$,

$X_{\text{maturation},k} \sim \text{Bin}\left(J_k(t), 1 - e^{-\gamma\Delta t}\right)$, $X_{\text{death},k} \sim \text{Bin}\left(J_k(t), 1 - e^{-(\mu_J + \alpha\rho_k)\Delta t}\right)$,

where $F_G^{(k)}$ is the number of gravid females located within the oviposition radius of breeding site s_k , and the local density term is defined as $\rho_k = \frac{J_k(t)}{K_c A_d}$, with $A_d = (2r_d + 1)^2$ the area of the neighborhood used to compute density-dependent mortality.

Adult males die with rate μ_M , while females die with rate μ_F . Unmated females become gravid after encountering males within mating radius r_m . If $M_r(x)$ denotes the number of nearby males, the mating probability during Δt is $P_{\text{mating}} = 1 - e^{-\beta M_r(x)\Delta t}$.

Gravid females oviposit with rate f , producing juveniles at the nearest breeding site when located within oviposition radius r_o . Attraction toward breeding sites is modeled through a spatial potential field

$$\Pi(y) = \sum_{b \in \mathcal{S}} \exp\left(-\frac{\|y - b\|_2}{\lambda}\right), \quad (2)$$

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where λ controls the spatial decay. Adult movement is represented as a stochastic lattice jump. From position x , the next position is sampled from $N_R(x) = \{y \in \mathcal{G} \mid \|y - x\|_2 \leq R, c(y) \notin C_{\text{forbidden}}\}$. Transition probabilities follow

$$P(y|x) = \left(\frac{\Pi(y) + \beta_0}{\|y - x\|_2 + 1} \right) / \left(\sum_{z \in N_R(x)} \frac{\Pi(z) + \beta_0}{\|z - x\|_2 + 1} \right), \quad (3)$$

where β_0 controls exploratory movement. This transition kernel defines a Markov.

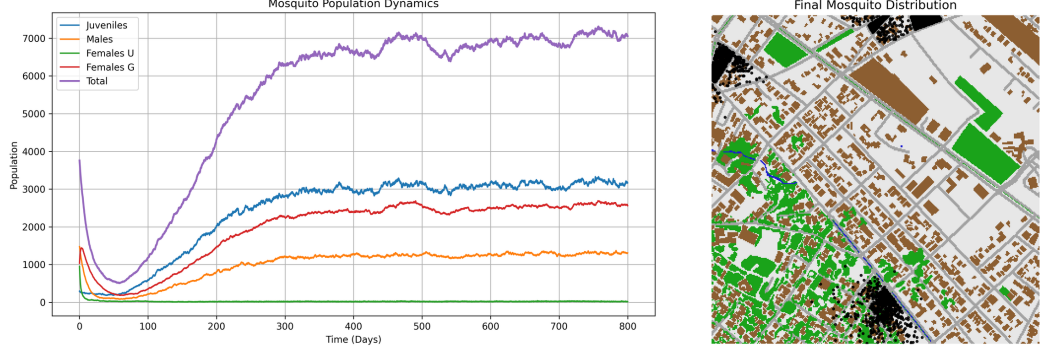


Figure 1: Results of a Simulation. Parameters: Initial population: 800 juveniles, 1000 adult males, 1000 unmated females, and 1000 gravid females. Biological rates (day^{-1}): $\mu_J = 0.05$, $\mu_M = 0.10$, $\mu_F = 0.08$, $\gamma = 0.07$, $\alpha = 0.001$, $K_c = 100$, $\beta = 0.02$, $f = 5.0$. Movement parameters: jump radius $R = 3$, exploration bias $\beta_0 = 10$. Spatial interaction radius (cells): $r_m = 5$, $r_d = 3$, $r_o = 10$. Time step $\Delta t = 1.0$ day. Source: Own elaboration.

This model combines spatial heterogeneity, stochastic interactions, and probabilistic movement consistent with reaction–diffusion dynamics. It enables the analysis of how breeding site distribution and urban structure influence mosquito populations and can generate synthetic in–silico data for parameter calibration and hybrid data–driven models [1].

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